

Executive Summary

Factory Reallocation & Shipping Optimization — Nassau Candy Distributor

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The Problem

Nassau Candy currently assigns each product to a factory using fixed, static rules that were never designed to account for actual shipping distance or demand patterns. Our analysis of 10,194 real orders shows this is costing the business in longer delivery times and avoidable logistics inefficiency — particularly for products shipped out of the Sugar Shack facility.

What We Built

A predictive analytics system that:

- Predicts how long a shipment will take based on product, factory, destination, and shipping method — accurate to within about 7.5 hours, on average.
- Automatically flags which factory-to-region shipping routes are chronically slow or overloaded.
- Simulates what would happen if any product were made at a different factory instead — before making any real change.
- Recommends the best factory for every product, ranked by expected time savings.

All of this is available through an interactive dashboard where any team member can select a product and instantly see the recommended factory and expected improvement.

What We Found

Metric	Result
Products analyzed	15 (100% of catalog)
Products with a faster alternative factory	12 of 15 (80%)
Best possible improvement found	13.8% faster delivery
High-risk / congested shipping routes identified	15 routes
Prediction accuracy of the model	93.5% (highly reliable)

Top Opportunities

Four products currently made at Sugar Shack — Laffy Taffy, Nerds, SweetARTS, and Fizzy Lifting Drinks — would ship noticeably faster (8–14% improvement) if reassigned to Secret Factory or The Other Factory. Two Wicked Choccy's chocolate products bound for the Pacific region would also benefit from moving to Secret Factory.

Recommended Next Steps

- Review the top reassignment opportunities above with the operations team for feasibility (capacity, contracts, transition cost).
- Treat the 15 flagged high-risk routes as the priority list for any near-term logistics investment.
- Pilot one or two low-risk reassignments first and measure real-world impact before a full rollout.
- Feed real shipment tracking and freight cost data back into the model over time to further improve accuracy.

Bottom Line

Without changing prices, staffing, or products, Nassau Candy has a clear, data-backed opportunity to reduce delivery times for the majority of its catalog simply by reassigning where products are made — and now has the tooling to evaluate any future change before committing to it.